



told us about the hardware that Driveri uses. Having closely worked on the device since its inception, he describes the product as a high performing road safety and driver monitoring device that is equipped with advanced sensor processing and provides a 360-degree, high-definition video recording. To achieve this level of high functionality, the product has gone through a series of quality checks and rigorous transformation. “When a device records around 30 frames per second, it uses a very powerful processor that can run all these vision-based algorithms on the device in real-time. We have used a high-power GPU to provide a high-level overview,” stated a delighted Teja Gudena.

With such high performing power, challenges are plenty and the team is proud to have overcome them all. Gudena recalled that since the device functions with high processing power, the team encountered challenges in terms of power, form factor, and even thermal impact but the first prototype came out for testing a year later and there has been no going back on the core system. As with the case of any other

device, there have been upgrades to the existing design, but there have been no changes to the hardware concept.

While Driveri was initially designed to suit the climatic conditions of North America, it has seen a major transformation in terms of adaptability to other regions as well. The device’s accuracy remains unaffected even in conditions that are hot and humid, despite its own thermal dissipation. Driveri’s advanced machine learning allows it constantly to learn new information and be able to adapt to the region’s signboards, metric systems, and much more. Teja Gudena stated that the capability of Driveri is far more advanced than its competitors simply because of its high processing power and algorithms and its ability to send only meaningful data to the cloud. “The device practically works like a human being monitoring the entire drive and alerting the driver, which requires high levels of algorithms. We have used such advanced technology in Driveri long before any of our competitors had even thought of it,” quipped Gudena.

With a steadfast approach to releasing quality products into the market, Netradyne believes that there just cannot be any room for any kind of failure in the system. The device is designed to recover from failures and continue to function to its optimum capacity. To negate dubieties, Driveri even endures a series of negative testing before it hits the market. For a device that practically fits right in your hand, the sheer volume of data it records, analyses, interprets and warns about is nothing short of breathtaking. Senior Vice President, Cloud Engineering at Netradyne, Vinay Rai, explains the scalability of the product. Netradyne’s product has processed and analysed over 15 billion minutes of drive time data and continues to collect over 1 billion minutes every single month. “This highlights how much training our system has the advantage of learning from, thereby making it more resilient,” remarked Rai.

Driveri has been designed to send only meaningful data to cloud to ensure users only get the actionable insights. However, Vinay Rai pointed out that the device provides a video for every alert sent out and



Features	Driver.i D210	Driver.i D215
Deep Learning Processor	Qualcomm	Qualcomm
Storage	Up to 100 hrs of drive-time	Up to 240 hrs of drive-time
Cameras	Dual-HD setup	Dual-HD setup
Communication Channels	External 4G-LTE / Wi-Fi / Bluetooth / GPS module Integrated with CAN Bus (J1939 / OBDII)	Integrated 4G-LTE / Wi-Fi / Bluetooth / GPS module Integrated with CAN Bus (J1939 / OBDII)
Inertial Sensors	Accelerometer, Gyroscope and Magnetometer	Accelerometer, Gyroscope and Magnetometer

Driveri™ Vision-based Driver Safety System